MTH 206-INTRODUCTION TO NUMERICAL ANALYSIS

CONTENT

MODULE 1-APPROXIMATIO: Least Squares Polynomial Approximation

Chebychev's Polynomial.

Continued Fraction and Rational Function

Orthogonal Polynomial.

Numerical Integration Methods (Trapezium and Simpson's Rules, Newton-Cotes, Gaussian).

Matrices: Eigenvalues and Eigenvector

Least Squares Approximation

- Often it may be difficult to find a function y = y(x) except by certain techniques. One of the known methods of fitting a polynomial function to this set of data is the Least squares approach. The least squares approach is a technique which is developed to reduce the sum of squares of errors in fitting the unknown function.
- The Least Squares Approximation methods can be classified into two, namely

  i) **Discrete least square approximation:** involves fitting a polynomial function to a set of data points using the least squares approach,

  ii) **Continuous least squares approximation:** requires the use of orthogonal polynomials to determine an appropriate polynomial function that fits a given function.

Discrete Least Squares Approximation

The basic idea of least square approximation is to fit a polynomial function P(x) to a set of data points (X_i, Y_i) having a theoretical solution.

$$y = f(x) \qquad (1.1)$$

The aim is to minimize the squares of the errors. In order to do this, suppose the set of data satisfying the theoretical solution (1.1) are

$(X_1, Y_1), (X_2, Y_2) \ldots, (X_n, Y_n)$

Attempt will be made to fit a polynomial using these set of data points to approximate the theoretical solution $f(x)$.

The polynomial to be fitted to these set of points will be denoted by $\mathrm{P}(x)$ or sometimes $P_n(x)$ to denote a polynomial of degree n. The curve or line $\mathrm{P}(x)$ fitted to the observation $y_1, y_2, \ldots, y_n$ will be regarded as the best fit to $f(x)$, if the difference between $\mathrm{P}(x_i)$ and $f(x_i)$, $i = 1, 2, \ldots, n$ is least. That is, the sum of the differences

$e_i = f(x_i) - \mathrm{P}(x_i)$, $i = 1, 2, \ldots, n$ should be the minimum.

- The differences obtained from e_i could be negative or positive and when all these e_i are summed up, the sum may add up to zero. This will not give the true error of the approximating polynomial. Thus to estimate the exact error sum, the square of these differences are more appropriate. In other words, we usually consider the sum of the squares of the deviations to get the best fitted curve.

Thus the required equation for the sum of squares error is then written as

$$S = \sum_{i=1}^{n} \left[f(x_i) - P(x_i) \right]^2 \qquad 1.2$$

which will be minimized.

where P(x) is given by

$$P(x) = a_0 + a_1 x + a_2 x^2 + \ldots + a_n x^n \qquad 1.3$$

The above approach can be viewed either in the discrete case or in the continuous case

Fitting of polynomials (Discrete case)

We shall now derive the formula in discrete form that fits a set of data point by Least squares technique. The aim of least squares method is to minimize the error of squares.

To do this we begin by substituting equations (1.1) and (1.3) in equation (1.2), this gives:

$$S = \sum_{i=1}^{n} \left[y_i - (a_o + a_1 x_i + a_2 x_i^2 + \ldots + a_k x_i^k) \right]^2 \qquad 1.4$$

To minimize S, we must differentiate S with respect to a_i and equate to zero. Hence, if we differentiate equation (1.4) partially with respect to a_0, a_1, …, a_k, and equate each to zero, we shall obtain the following:

$$\left.\begin{aligned}
\frac{\partial S}{\partial a_o} &= -2\sum_{i=1}^{n}\left[y_i - (a_o + a_1 x_i + a_2 x_i^2 + \ldots + a_k x_i^k)\right] = 0 \\
\frac{\partial S}{\partial a_1} &= -2\sum_{i=1}^{n}\left[y_i - (a_o + a_1 x_i + a_2 x_i^2 + \ldots + a_k x_i^k)\right] x_i = 0 \\
\frac{\partial S}{\partial a_2} &= -2\sum_{i=1}^{n}\left[y_i - (a_o + a_1 x_i + a_2 x_i^2 + \ldots + a_k x_i^k)\right] . x_i^2 = 0 \\
&\ldots\ldots\ldots\ldots \\
\frac{\partial S}{\partial a_k} &= -2\sum_{i=1}^{n}\left[y_i - (a_o + a_1 x_i + a_2 x_i^2 + \ldots + a_k x_i^k)\right] . x_i^k = 0
\end{aligned}\right\} \qquad 1.5$$

These can be written as

$$\left.\begin{aligned}
\sum y_i &= na_o + a_1\sum x_i + a_2\sum x_i^2 + \ldots + a_k\sum x_i^k \\
\sum x_i y_i &= a_o\sum x_i + a_1\sum x_i^2 + a_2\sum x_i^3 + \ldots + a_k\sum x_i^{k+1} \\
\sum x_i^2 y_i &= a_o\sum x_i^2 + a_1\sum x_i^3 + a_2\sum x_i^4 + \ldots + a_k\sum x_i^{k+2} \\
&\ldots\ldots\ldots \\
\sum x_i^k y_i &= a_o\sum x_i^k + a_1\sum x_i^{k+1} + a_2\sum x_i^{k+2} + \ldots + a_k\sum x_i^{2k}
\end{aligned}\right\} \quad 1.6$$

where $\sum$ is assumed as short form for $\sum_{i=1}^{n}$

Solving equation (1.6) to determine $a_0,\ a_1, \ldots, a_k$ and substituting into equation (1.3) gives the best fitted curve to (1.1).

The set of equations in (1.6) are called the **Normal Equations** of the Least Squares Method

Equation (1.6) can only be used by creating a table of values corresponding to each sum and the sum is found for each summation.

We shall now illustrate how to use the set of equations (1.6) in a tabular form.

Example 1

By using the Least Squares Approximation, fit

(a) a straight line

(b) a parabola

to the given data below

x	1	2	3	4	5	6
y	120	90	60	70	35	11

Which of these two approximations has least error?

Solution

(a) In order to fit a straight line to the set of data above, we assume the equation of the form

$$y = a_o + a_1 x$$

The graph of the set of data above is given by in figure 1

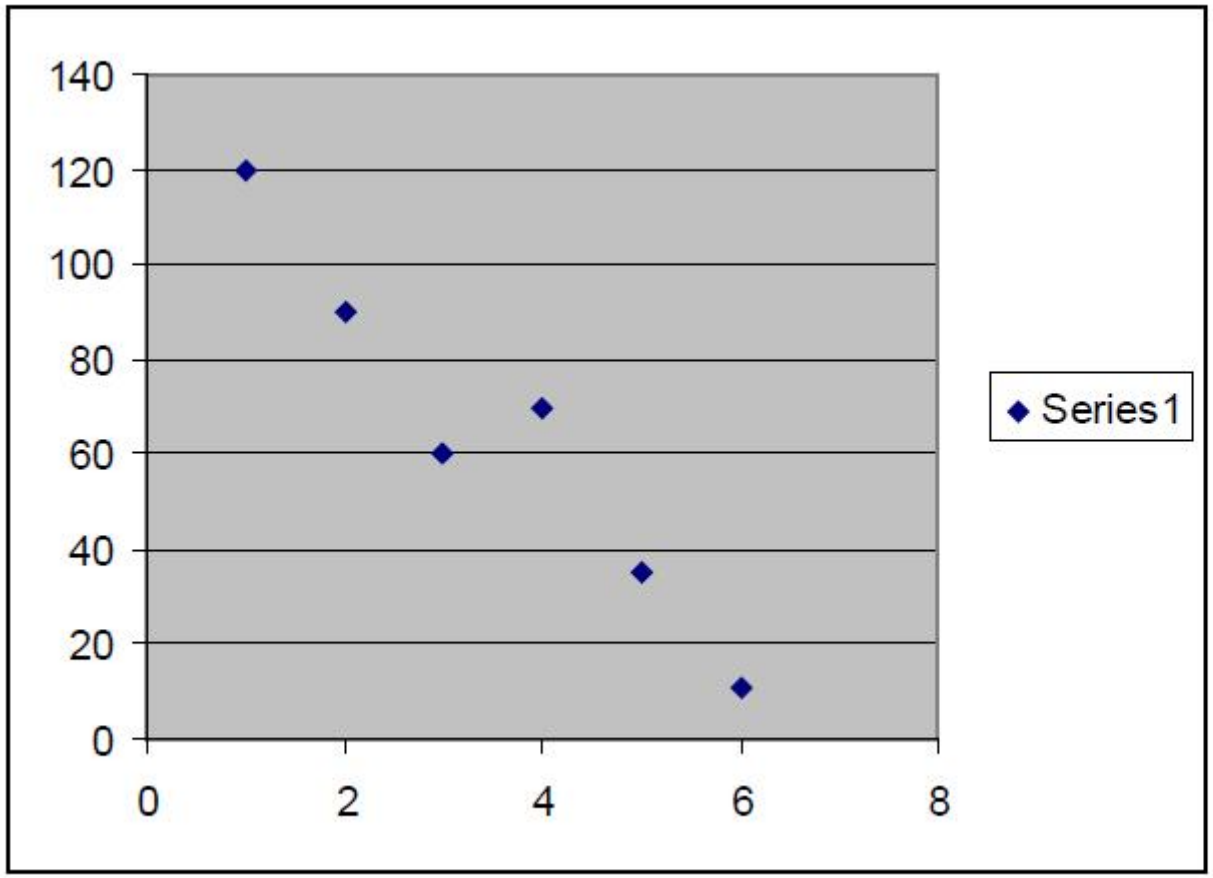

Figure 1

By inspection a straight line may be fitted to this set of data as the line of best fit, since most of the points will lie on the fitted line or close to it. However, some may want to fit a curve to this but the accuracy of the curve fitted is a thing for consideration. Now from the straight line equation above, we have to determine two unknowns a_0 *and* a_1, the normal equations necessary to determine these unknowns can be obtained from equation (1.6) as:

$$\sum y_i = na_o + a_1 \sum x_i$$

$$\sum x_i y_i = a_o \sum x_i + a_1 \sum x_i^2$$

Hence we shall need to construct columns for vales XY and x^2 in addition to X and Y values already given. Thus the table below shows the necessary columns:

Table 1

x	y	x^2	xy
1	120	1	120
2	90	4	180
3	60	9	180
4	70	16	280
5	35	25	175
6	11	36	66
Σ 21	**386**	**91**	**1001**

$$386 = 6a_o + 21a_1$$

$$1001 = 21a_o + 91a_1$$

Solving these two equations, we obtain

$$a_o = 134.33\ , \quad a_1 = -20$$

Therefore the straight line fitted to the given data is

$$y = 134.33 - 20x$$

(b) In a similar manner, the parabola can be written as $y = a_o + a_1 x + a_2 x^2$. Hence the required normal equations to determine the unknowns a_o, a_1 and a_2 are:

$$\sum y_i = n a_o + a_1 \sum x_i + a_2 \sum x_i^2$$

$$\sum x_i y_i = a_o \sum x_i + a_1 \sum x_i^2 + a_2 \sum x_i^3$$

$$\sum x_i^2 y_i = a_o \sum x_i^2 + a_1 \sum x_i^3 + a_2 \sum x_i^4$$

Thus, we construct the necessary columns in Table 2 given below:

Table 2

x	y	x^2	x^3	x^4	xy	x^2y
1	120	1	1	1	120	120
2	90	4	8	16	180	360
3	60	9	27	81	180	540
4	70	16	64	256	280	1120
5	35	25	125	625	175	875
6	11	36	216	1296	66	396
21	**386**	**91**	**441**	**2275**	**1001**	**3411**

Substituting into the normal equations, we have

$$6a_o + 21a_1 + 91a_2 = 386$$

$$21a_o + 91a_1 + 441a_2 = 1001$$

$$91a_o + 441a_1 + 2275a_2 = 3411$$

On carefully solving these equations, we obtain

$$a_o = 136\,, \quad a_1 = -\frac{85}{4}\,, \quad a_2 = \frac{5}{28}$$

As a learner, you are expected to solve these equations by any simple method known to you.

Hence the fitted curve is $y = 136 - \frac{85}{4}x + \frac{5}{28}x^2$

The graphs of the two approximate functions are shown below:

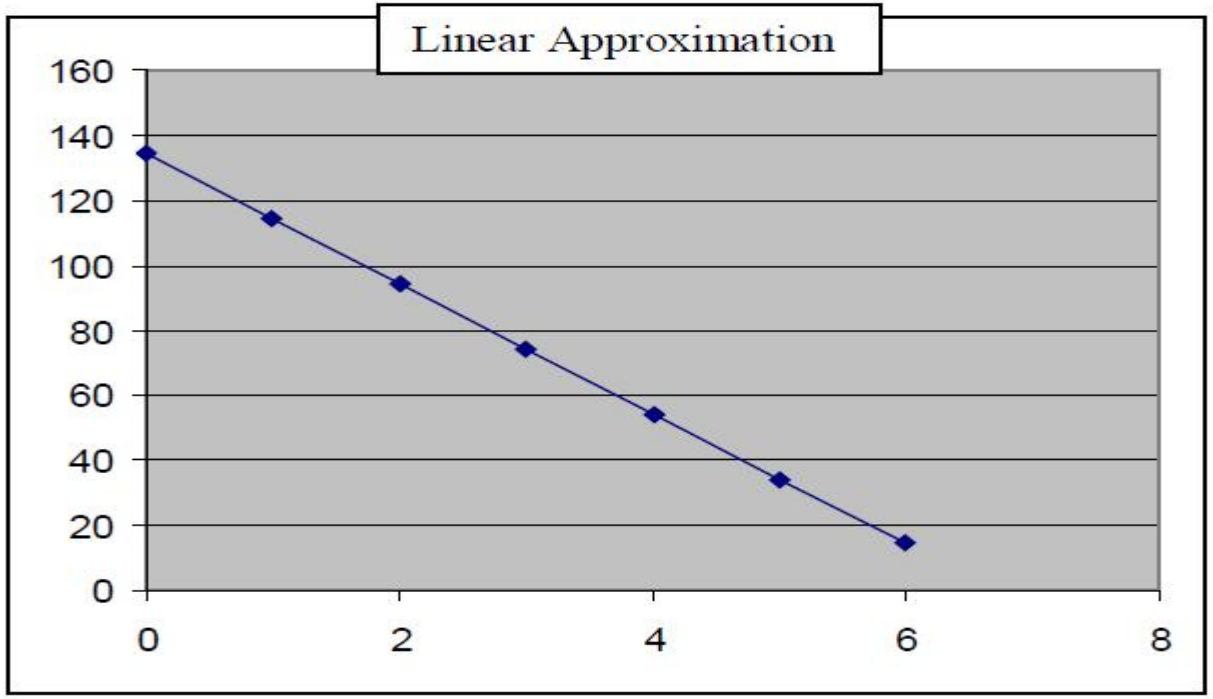

Figure 2

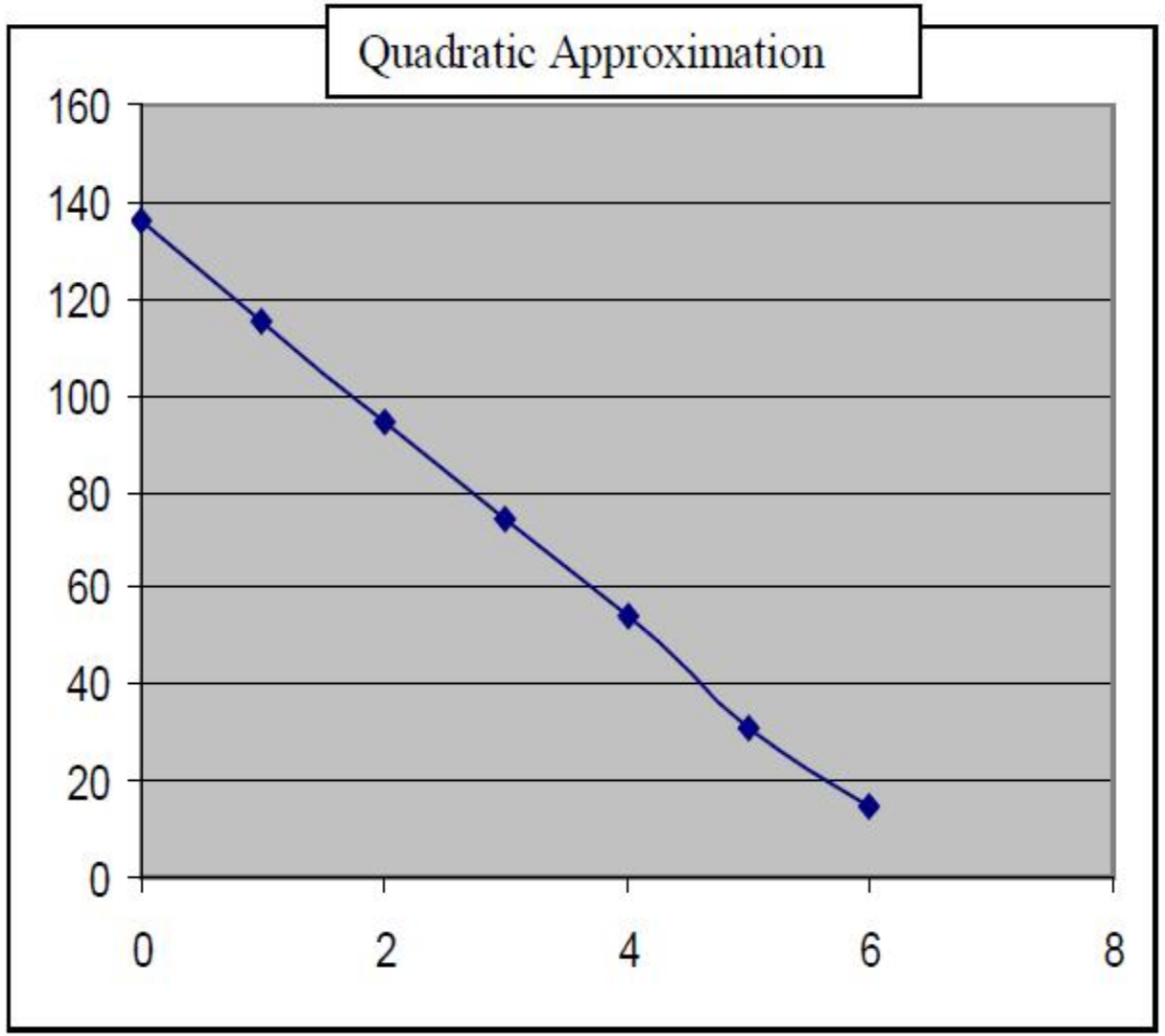

Figure 3

The two lines look like a straight line but the errors of the two lines will depict the better approximation.

We can now consider the accuracy of the two fitted functions to the data and decide on which one of the two approximations is better. To do this we may retain values of x for 1, 2, . . . , 6 and evaluate the corresponding y values.

For example when x = 1, the linear approximation gives y = 134.33 – 20 = 114.33, where as for the same x = 1 the quadratic approximation gives: $y = 136 - \frac{85}{4} + \frac{5}{28} = 114.9$. For the set of values of x tabulated we have the corresponding values for y in both approximations. The squares of the errors are considered as $(y - y_i)^2$. The table below describes the error table.

Table 3

x	y	$y_1 = y(L)$	$y_2 = y(Q)$	$(y-y_1)^2$	$(y-y_2)^2$
1	120	114.33	114.9	32.1489	26.01
2	90	94.33	94.2	18.7489	17.64
3	60	74.3	73.9	204.49	193.21
4	70	54.3	53.9	246.49	259.21
5	35	34.3	31.3	0.49	13.69
6	11	14.3	14.9	10.89	15.21
				513.2578	524.97

The sums of squares of the error of fitted lines by linear function y(L) and quadratic function y(Q) are shown above in Table 3. The comment here is that the sum of squares of error of the linear (513.26) is slightly lower than that of the quadratic (524.97). Invariably the linear function is a better approximation to the data above.

Conclusion

Fitting a polynomial to discrete data by least squares methods is easily handled by creating tables of values and generating all necessary columns that will enable one to obtain the normal equations. The normal equations are then solved simultaneously to determine the unknowns which are then substituted into the required approximate polynomial equation.

ASSIGNMENT

1. The table below gives the readings from a laboratory experiment.

Time t	2	3	5	6	9
Reading y	7	17	49	71	161

Fit (a) a linear function and (b) a quadratic polynomial to the above data by method of least squares and determine which of the two is a better approximation.

2. How many normal equations will be needed to fit a cubic polynomial? Hence list the entire necessary variables for such fitting.

2.0 Least Squares Approximation (Continuous Case)

Introduction

We have seen in the last unit how to fit a polynomial to a set of data points by using the least squares approximation technique. We shall in this unit consider the situation in which a function f(x) is to be approximated by the least squares method in terms of a polynomial. In this case since data are no longer given, it would not be necessary to create tables by columns but rather by carrying out some integration. The candidate is therefore required to be able to integrate simple functions as may be required.

Fitting an Approximate Polynomials (Continuous case)

- If we wish to find a least square approximation to a continuous function f(x), our previous approach must be modified since the number of points (x_i, y_i) at which the approximation is to be measured is now infinite (and non-countable). Therefore, we cannot use a summation as

$$\sum_{i=1}^{n}[f(x_i)-P(x_i)]^2$$

but we must use a continuous measure, that is an integral. Hence if the interval of the approximation is [a, b], so that $a \le x \le b$ for all points under consideration, then we must minimize

$$\int_a^b [f(x)-P(x)]^2 \, dx$$

where y = f(x) is our continuous function and P(x) is our approximating function.

Derivation

Let f(x) be a continuous function which in the interval (a, b) is to be approximated by a polynomial linear combination

$$P(x) = a_o + a_1 x + a_2 x^2 + \ldots + a_k x^k \qquad 2.1$$

$$S = \int_a^b \left[y_i - (a_o + a_1 x_i + a_2 x_i^2 + \ldots + a_k x_i^k)\right]^2 dx$$

of n+1 given functions $\varphi_o, \varphi_1, \varphi_2 \ldots, \varphi_n$. Then, $c_0, c_1, \ldots, c_n$ can be determined such that a weighted Euclidean norm of the error function f(x) – p(x) becomes as small as possible

That is, $$\|f(x)-P(x)\|^2 = \int_a^b |f(x)-P(x)|^2 \, w(x)\,dx \qquad 2.2$$

where w(x) is a non-negative weighting function. Equation (2.2) is the continuous least square approximation problem.

The minimization problem of f(x) by continuous function P(x) of (2.1) is given by

$$S = \int_a^b [f(x)-P(x)]^2 \, dx \qquad 2.3$$

where the interval [a, b] is usually normalized to [-1,1] following Legendre polynomial or Chebyshev function approximation, hence, substituting equation (2.2) in (2.3), we obtain

$$S = \int_{-1}^{1} [f(x)-\{c_o\varphi_o(x)+c_1\varphi_1(x)+c_2\varphi_2(x)+\ldots+c_n\varphi_n(x)\}]^2 \, w(x)\,dx \qquad 2.4$$

where $\varphi_o, \varphi_1, \varphi_2 . \ldots, \varphi_n$ are some chosen polynomials.

To minimize (3.3) or (3.4), one could consider two alternative methods to obtain the coefficients or the unknown terms.

First Method

The first approach for minimizing equation (3.3) is by carrying out an expansion of the term $[f(x) - P(x)]^2$, next carry out the integration and then by means of calculus, obtain the minimum by setting

$$\frac{\partial S}{\partial c_k} = 0 \quad , \quad k = 0, 1, 2, \ldots, n$$

This approach will be illustrated by the next example.

Example

Find the least square straight line that provides the best fit to the curve $y = \sqrt{x}$ over the interval $0 \le x \le 1$.

Solution

Let the line be y = ax+b, we must minimize

$$S = \int_0^1 \left[\sqrt{x} - ax - b\right]^2 \, dx$$

Expand the integrand, we obtain

$$S = \int_0^1 \left[x - 2ax^{3/2} - 2b\sqrt{x} + a^2x^2 + 2abx + b^2\right] dx$$

And integrating we get

$$S = \left[\tfrac{1}{2}x^2 - \tfrac{4}{5}ax^{5/2} - \tfrac{4}{3}bx^{3/2} + \tfrac{a^2}{3}x^2 + abx^2 + b^2x\right]_0^1$$

Evaluating we get

$$S = \frac{a^2}{3} + b^2 + ab - \frac{4}{5}a - \frac{4}{3}b + \frac{1}{2}$$

For a minimum error, we must set $\dfrac{\partial S}{\partial a} = 0$, *and* $\dfrac{\partial S}{\partial b} = 0$

Doing this, we get

$$\frac{\partial S}{\partial a} = \frac{2}{3}a + b - \frac{4}{5} = 0$$

$$\frac{\partial S}{\partial b} = 2b + a - \frac{4}{3} = 0$$

Thus, we solve the equations

$$\frac{2}{3}a + b = \frac{4}{5}$$

$$a + 2b = \frac{4}{3}$$

Solving, we get $a = \dfrac{4}{5}$, $b = \dfrac{4}{15}$

Hence the Least squares approximation is

$$y = \frac{4}{5}x + \frac{4}{15}$$

WE observe that The line $y = \frac{4}{5}x + \frac{4}{15}$ meets the curve $y = \sqrt{x}$ in two points P(0.1487, 0.3856) and Q(0.7471, 0.8643) as shown below.

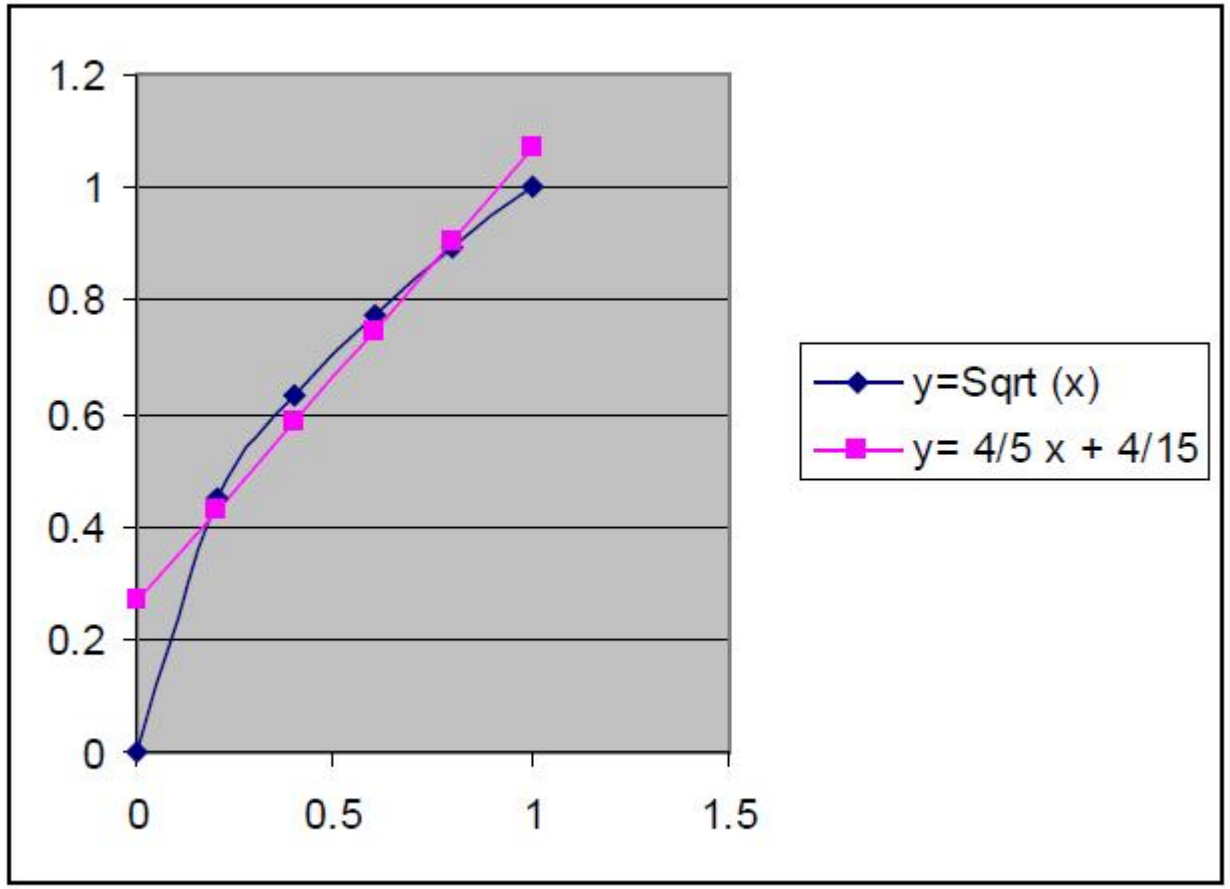

Figure 1

This straight line is only a linear approximation to the given curve. We could have as well found other polynomial approximation using the same least squares technique.

It will be observed that if P(x) is a polynomial of higher degree say n = 3 or more, the expression $[f(x) - P(x)]^2$ may not be easy to expand before integrating, so we must seek another approach for the minimization.

Alternative Method

Now, suppose we wish to find the least squares approximation, using a polynomial of degree k to a continuous function y over [a,b]. In such a case, we must minimize the integral

$$S = \int_a^b \left[y_i - (a_o + a_1 x_i + a_2 x_i^2 + \ldots + a_k x_i^k) \right]^2 \, dx$$

If we do not want to expand the expression in the squared bracket, then we must first get the normal equations. In other words, we derive the normal equation by obtaining $\frac{\partial S}{\partial a_i} = 0$, before integrating the resulting function and evaluating the result.

Doing this we obtain

$$\frac{\partial S}{\partial a_o} = \int_a^b -2\left(y_i - a_o - a_1 x - a_2 x^2 - \ldots - a_k x^k\right) dx = 0$$

$$\frac{\partial S}{\partial a_1} = \int_a^b -2x\left(y_i - a_o - a_1 x - a_2 x^2 - \ldots - a_k x^k\right) dx = 0$$

$$\frac{\partial S}{\partial a_2} = \int_a^b -2x^2\left(y_i - a_o - a_1 x - a_2 x^2 - \ldots - a_k x^k\right) dx = 0$$

and in general we can write

$$\frac{\partial S}{\partial a_r} = \int_a^b -2x^r\left(y_i - a_o - a_1 x - a_2 x^2 - \ldots - a_k x^k\right) dx = 0 \quad \text{for} \quad (r = 0, 1, \ldots, k)$$

The factor (–2) that appeared first in the integral can be ignored since the right hand side of the equation is zero. Hence, the normal equations can be written as

$$\int_a^b x^r\left(y_i - a_o - a_1 x - a_2 x^2 - \ldots - a_k x^k\right) dx = 0 \qquad \text{for} \quad (r = 0, 1, \ldots, k)$$

This will give (k+1) linear equations in the (k+1) unknowns $a_0, a_1, \ldots, a_k$ which can be solved simultaneously by any algebraic process.

This approach may be simpler than the first one and we therefore suggest this second method. However any of the two techniques may be used and are both valid.

Example 2

Find the least squares quadratic $ax^2 + bx + c$, which best fits the curve $y = \sqrt{x}$ over the interval $0 \le x \le 1$.

Solution

We need to minimize

$$S = \int_0^1 \left[\sqrt{x} - ax^2 - bx - c\right]^2 dx$$

By this new approach, we shall first of all obtain the normal equations. Thus we have:

$$\frac{\partial S}{\partial c} = \int_0^1 \left(\sqrt{x} - ax^2 - bx - c\right) dx = 0$$

$$\frac{\partial S}{\partial b} = \int_0^1 x\left(\sqrt{x} - ax^2 - bx - c\right) dx = 0$$

$$\frac{\partial S}{\partial a} = \int_0^1 x^2\left(\sqrt{x} - ax^2 - bx - c\right) dx = 0$$

Integrating, we get the three equations as follows:

$$\frac{2}{3}x^{3/2} - \frac{1}{3}ax^3 - \frac{1}{2}bx^2 - cx\Big|_0^1 = 0$$

$$\frac{2}{5}x^{5/2} - \frac{1}{4}ax^4 - \frac{1}{3}bx^3 - \frac{1}{2}cx^2\Big|_0^1 = 0$$

$$\frac{2}{7}x^{7/2} - \frac{1}{5}ax^5 - \frac{1}{4}bx^4 - \frac{1}{3}cx^3\Big|_0^1 = 0$$

Evaluating within the limits we obtain three simultaneous equations

$$\frac{1}{3}a+\frac{1}{2}b+c=\frac{2}{3}$$
$$\frac{1}{4}a+\frac{1}{3}b+\frac{1}{2}c=\frac{2}{5}$$
$$\frac{1}{5}a+\frac{1}{4}b+\frac{1}{3}c=\frac{2}{7}$$

Solving these equations simultaneously we get

$$a=-\frac{4}{7}\ ,\quad b=\frac{48}{35}\ ,\quad c=\frac{6}{35}$$

Thus the least squares quadratic function is

$$f(x)=-\frac{4}{7}x^2+\frac{48}{35}x+\frac{6}{35}$$

or $$f(x)=-\frac{2}{35}\left(10x^2-24x-3\right)$$

The earlier approach of expanding the integrand is also valid. However the number of terms in the expansion may not be friendly. The higher the degree of the polynomial we are trying to fit by least squares approximation, the more difficult it is to obtain the coefficients of the polynomial by expansion. Hence this last approach or method may be more appropriate for easy handling.

The error table of the above result is given below

x	$y=\sqrt{x}$	$f(x)=-\frac{2}{35}\left(10x^2-24x-3\right)$	E= | y – f(x) |
0	0.0000	0.1714	0.1714
0.1	0.316228	0.302857	0.013371
0.2	0.447214	0.422857	0.024357
0.3	0.547723	0.531429	0.016294
0.4	0.632456	0.628571	0.003885
0.5	0.707107	0.714286	0.007179
0.6	0.774597	0.788571	0.013974
0.7	0.836660	0.851429	0.13666
0.8	0.894427	0.902857	0.094427
0.9	0.948683	0.942857	0.048683
1.0	1.0000	0.971429	0.0000

Conclusion

To conclude this unit, it would be observed using the least squares approximation technique to fit a polynomial to a continuous function f(x) could be done in two ways. You are allowed to use either of the methods. However, expanding the integrand before carrying out the integration may be sometimes more difficult than to first of all find the normal equations. The error obtained in using the least square method will depend on how well a polynomial of certain degree is suitable in approximating the continuous function.